

CO1 - ASSESSMENT TOOL - I

SHORT ANSWER

1. Define cloud computing and explain any four characteristics that makes it different from traditional on-premise computing.

Cloud computing is the on demand delivery of computing services including servers, storage, databases, networking and software over the internet with pay-as-you-go pricing.

characteristics vs on premise:

On demand self service: Users can provision computing capabilities automatically without requiring human interaction with the service provider.

Elasticity and scalability: Resource can be scaled up or down rapidly based on demand: on premise requires manual hardware upgrades.

Pay-as-you-go Pricing: You pay only for what you use (OPEX) whereas on-premise requires significant upfront capital investment.

Broad Network Access: Services are available over the network and accessed through standard mechanisms.

2. **True Evolution of Cloud Computing Hardware:**

Hardware Evolution:

Large mainframe computers were used in organizations.

Personal Computers: Affordable PCs made computing accessible to individuals.

Client-Server Architecture: Clients requested services from centralized servers.

Virtualization: Multiple virtual machines ran on a single physical server, improving efficiency.

Internet Evolution: High speed internet enabled remote access to applications and data.

Web Applications: Software became available through web browsers.

cloud computing: Providers like AWS, Microsoft Azure, and Google cloud began offering scalable service over the Internet.

3. Explain the role of server virtualization:

Role of Server Virtualization:

★ Divides one physical server into multiple Virtual Machines

★ Improves hardware utilization

★ Reduces infrastructure cost

★ Enables quick deployment and easy

Management of resources.

Virtualization	Cloud Computing
→ Technology that creates virtual or virtual machines	→ Service model that delivers computing resources over the internet
→ Focuses on efficient use of hardware	→ Focuses on providing service to user
→ Runs on a physical server	Uses virtualization to provide scalable services
Eg: VMware, Hyper-V	Eg: Microsoft Azure, Google Cloud

Discusses the role of data centres in delivering cloud service:

Data centres are facilities that contain servers, storage device, networking equipment and cooling system used to provide cloud service.

Roles:

- * Store and process user data
- * Host cloud computing and website
- * Provide secure backup and disaster recovery.
- * Ensure high availability and reliability.
- * Enable virtualization and resource sharing.

* Deliver cloud services to user world wide through the internet.

5.

Differentiation IaaS, PaaS, SaaS & XaaS with one suitable real world example:

Service Model	Description	Example
IaaS (Infrastructure as a Service)	Provides virtual servers storage and networking	Amazon EC2 (AWS)
PaaS (Platform as a Service)	Provides a platform to develop and deploy application	Google App (Engine)
SaaS (Software as a Service)	Delivers software application over the internet	Google Docs
XaaS (Anything)	Provide various IT services through the cloud	Microsoft 365